

Requirements engineering

it is the process of identifying, analyzing, specifying, validating & Managing the needs & expectation of stakeholders for a s/w System.

Steps:

1. Requirements elicitation: This is the process of gathering Information about the needs & expectations of stakeholders for the s/w System. This step involves interviews, Surveys, focus Groups and other techniques to gather Information from stakeholders.
2. Requirement Analysis: This step involves analyzing the Information gathered in the requirements elicitation step to identify the high level Goals and Objectives of the s/w System. It also involves identifying any Constraints or limitations that may affect the development of the s/w system.

3. Requirements Specification: This step involves documenting the requirements identified in the analysis step in a clear, consistent & unambiguous manner.

This step also involves prioritizing & grouping the requirements also manageable chunks.

4. Requirements Validation: This step involves checking that the requirements are complete, consistent & accurate. it also involves checking that the requirements are testable & that they meet the needs and expectations.

5. Requirements Management: This step involves managing the requirements throughout the SDLC including tracking & controlling changes, and ensuring that the requirements are still valid and relevant.

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Requirement engineering is the process of defining, documenting & Maintaining the requirements. It is a process of Gathering & defining service provided by system. It is the disciplined appⁿ of proven principle, Methods, tools and Notations to describe a proposed System's intended behaviour and its associated constraints.

Tools involved :

- ① Observation report
- ② Survey, Poll
- ③ Use Cases
- ④ User Stories
- ⑤ Requirement workshop
- ⑥ Mind Mapping
- ⑦ Role playing
- ⑧ Prototyping.

Requirements Engineering process

It is a critical step in SDLC as it helps to ensure that the s/w system being developed Meets the needs & expectations of stakeholders and that it is developed on time, within budget and to the required quality.

It Consists Main Activities :-

1. Requirements elicitation
2. Requirements specification
3. Requirements Verification & Validation
4. Requirements Management.

⇒ Requirement Elicitation

it is related to the various ways used to gain knowledge about the project and requirements.

The various sources include Customers, business Manuals, the existing s/w of same type, std. and other stakeholders of the project.

The techniques used are interviews, brainstorming, task analysis, Delphi technique, prototyping etc.

Requirement elicitation is the first step in the requirement engineering process and it is critical to the success of the s/w development project.

The Goal of this step is to understand the problem that the S/w system is to solve, & the needs & expectations of the stakeholders who will use the system.

1. Interview: These are one on one conversations to gather information about their needs & expectations.
2. Surveys: These are questionnaires that are distributed to gather information about their needs and expectations.
3. Focus Groups: These are small groups to discuss the needs & expectations for S/w system of customers.
4. Observation: This involves observing the stakeholders in their work environment to gather information about their needs and expectations.
5. Prototyping: This involves creating a working model of the S/w system, which can be used to gather feedback & to validate requirements.

It's Important to document, organize and prioritize the requirements obtained from all these techniques to ensure that they are complete, consistent & accurate.

⇒ Requirements Specification

This Activity is used to produce formal S/W requirement Models.

All the requirements including the functional as well as the non-functional requirements and the constraints are specified by these Models in totality.

During Specification, More knowledge about the problem may be required which can again trigger the elicitation process.

The Models used at this stage include ER diagrams, data flow diagrams (DFD), function decomposition diagrams (FDD), data dictionaries.

The Goal of this step is to create a clear and comprehensive document that describes the requirements for S/W system. This document should be understandable by both development team & the

Stake holders.

1. Functional Requirement : These describe what the s/w system should do. They specify the functionality that the system must provide such as Input Validation, data storage & user interface.
2. Non-functional Requirement : These describe how well the s/w system should do it. They specify the quality attributes of system i.e. performance, reliability, usability & security.
3. Constraints : These describe any limitations or restrictions that must be considered when developing the s/w system.
4. Acceptance Criteria : These describe the conditions that must be met for the s/w system to be considered complete & ready for release.

In order to make the requirements specification clear, the requirements should be written in a natural language & use simple terms,

and using a consistent format throughout the document.

it is also important to use diagrams, Models & other Visual aids to help communicate the requirements.

Once the requirements are specified, they must be reviewed & validated by the stakeholders & development team to ensure that they are complete, consistent & accurate.

⇒ Requirements Verification & Validation

Verification refers to the set of tasks that ensures that the S/W correctly implements a specific function.

Validation refers to a different set of tasks that ensures that the S/W that has been built is traceable to customer requirements. If requirements are not validated, errors in the requirement definitions would propagate to the successive stages resulting in a lot of modification & rework.

The Main Steps:

1. The requirement should be consistent with all the other requirements i.e. no two requirements should conflict with each other.
2. The requirements should be complete in every sense.
3. The requirements should be practically achievable.

Reviews, buddy checks, Making test cases, are some methods used for this.

Requirements Verification & Validation is the process of checking that the requirements for a software system are complete, consistent & accurate & that they meet the needs & expectation of stakeholders. The goal is to ensure that the software system being developed meets the requirements & that it is developed on time, within budget & to the required quality.

It is an iterative process that occurs throughout the SDLC.

It is Important to involve Stakeholders & the development team in this process to ensure that the requirements are thoroughly reviewed & -tested.

⇒ Requirements Management

It is a process of analyzing, documenting, tracking, prioritizing & agreeing on requirement & controlling the communication to relevant Stakeholders.

This stage takes care of changing Nature of requirements. It should be ensured that the SRS is as Modifiable as possible so as to incorporate changes in requirements specified by end users at later Stages too.

The Goal is to ensure that the System being developed Meets the Needs & expectation & that it is developed On-time, within budget & to the required quality.

Key activities:

1. Tracking & Controlling changes :

This involves Monitoring & Controlling changes to the requirements throughout the development process including identify the source of change, assessing the impact of change & approving or rejecting the change.

2. Version Control : This involves keeping track of different versions of requirements document & other related artifacts.

3. Traceability : This involves linking the requirements to other elements of the development process i.e design, testing & validation.

4. Communication : This involves ensuring that the requirements are communicated effectively to all stakeholders & that any changes or issues are addressed in a timely manner.

5. Monitoring & Reporting : This involves Monitoring the progress of development process & reporting on status of the requirements.

Advantages and Disadvantages of Requirements Engineering process.

Adv. : ① help to ensure that the s/w will Meets the Needs & expectations.

② identify problems early in the development process.

③ Improve Communication & Collaboration b/w Customer & client.

④ provide clear description of requirements to reduce errors.

⑤ helps to ensure that s/w System is delivered on time within a budget.

Disadv. : ① time Costly
② challenging
③ expensive process
④ May be complex Sometime
⑤ requirements may change over time, which can results in delays & additional costs.